

Vehicle Damage Assessment

Executive Summary · Deep-Learning Car Damage Classification

96.1%

Accuracy

0.961

F1

0.992

ROC-AUC

ResNet50

Best model

Objective

Automatically classify a car photo as **damaged** or **whole** to replace slow, inconsistent manual insurance inspections with an instant, auditable decision.

Approach

End-to-end pipeline on 2,300 balanced images: EDA → data cleaning → augmentation → three models (baseline CNN, ResNet50, EfficientNet-B0 transfer learning) → Optuna tuning → evaluation → Grad-CAM explainability → Streamlit app. Trained on an RTX 5080 (PyTorch, mixed precision).

Results (held-out validation, leak-free)

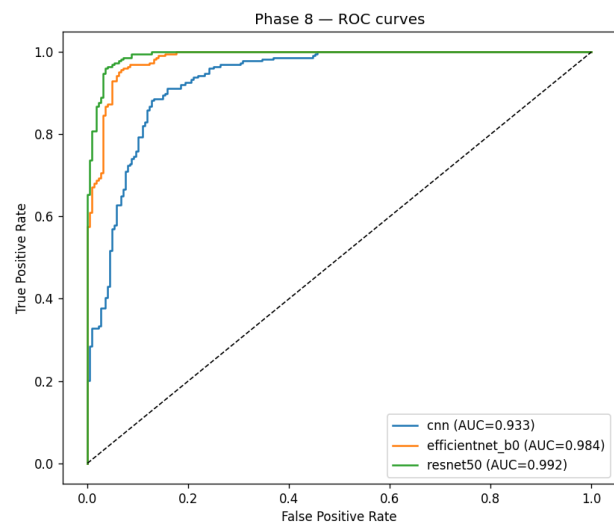
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline CNN	0.871	0.860	0.886	0.873	0.933
ResNet50	0.961	0.949	0.974	0.961	0.992
EfficientNet-B0	0.943	0.939	0.947	0.943	0.984

Key insight

EDA caught **data leakage** — identical images present in both the training and validation folders. Removing them keeps the reported metrics honest, a common pitfall that otherwise inflates accuracy.

Business value

	Manual	AI
Review time	Days	Seconds
Consistency	Varies	Deterministic
Fraud signal	Manual	Score + Grad-CAM
Scale	Headcount	1000s/hour



Recommendation: deploy ResNet50 as a first-pass triage classifier with adjuster review on low-confidence cases. Next steps: damage-severity grading and repair-cost estimation (require additional labels).